Engineering - Graduate Level Assessment 3 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. How much capacitance is required to resonate a coil of 0.001 mHat a frequency of 2 MHz?
   (a) 5,000 pF
   (b) 7,000 pF
   (c) 9,500 pF
   (d) 8,000 pF
   (e) 5,500 pF

2. A rectangular broad crested weir is 30 ft long and is known to have a discharge coefficient of 0.7. Determine the discharge if the upstream water level is 2 ft over the crest.
   (a) 465 cfs
   (b) 476 cfs
   (c) 490 cfs
   (d) 430 cfs
   (e) 450 cfs

3. Find the variance of the random variable X + b where X has variance,VarX and b is a constant.
   (a) 2 bE(X)
   (b) E(X^2) - [E(X)]^2
   (b) 2 VarX + b
   (c) VarX - b^2
   (d) VarX + 2 bVarX

4. f(X) = $[\pi(1 + X^2)]^{-}1 - \infty < x < \infty . IfY$ =X^2, what is the density function of Y ?
   (a) h(y) = [2 / {π(1 + $\sqrt{y}$)}] for y > 0 and = 0 otherwise
   (b) h(y) = [2 / {π(1 + y)}]y > 0 and = 0 otherwise
   (c) h(y) = [1 / {π(1 + y^2)}] for y > 0 and = 0 otherwise
   (d) h(y) = [1 / {2π(1 + $\sqrt{y}$)}] for y > 0 and = 0 otherwise
   (e) h(y) = [1 / {2π(1 + y^2)}] for y > 0 and = 0 otherwise

5. Calculate the endurance limit for a 1.75 in. AISI-C 1030 (as rolled) steel rod for 95% reliability.

(a) 24.2×10^3 psi
(b) 27.9×10^3 psi
(c) 18.5×10^3 psi
(d) 16.3×10^3 psi
(e) 22.8×10^3 psi

True / False (10 marks)

Answer True or False

6. True or False: A six-phase synchronous converter with a full-load d-c voltage of 600 volts produces an a-c voltage between rings of 300 volts.

7. True or False: Power dissipation in an ideal inductor is zero, as it does not consume energy but only stores and releases it.

8. True or False: Doubling the length of a cable will leave its capacitance C unchanged.

9. True or False: The variance of a random variable X uniformly distributed over the interval [0, 3] is 1/3.

10. True or False: The cutoff frequency of the first higher-order mode for the given coaxial line is 10 GHz.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the relationship between the preemphasis circuit characterized by the transfer function $|H_p(f)|^2 = k^2 f^2$ and the spectral density $G(f) = \frac{G_o}{1+(f/f_1)^2}$. Analyze how to determine an appropriate value for k^2 to ensure that preemphasis does not impact the bandwidth of the FM channel, considering the condition $f_1 << f_m$.

12. Discuss the bit requirements for storing a Binary-Coded Decimal (BCD) digit, analyzing the implications of this storage method on data representation in digital systems.

End of Paper